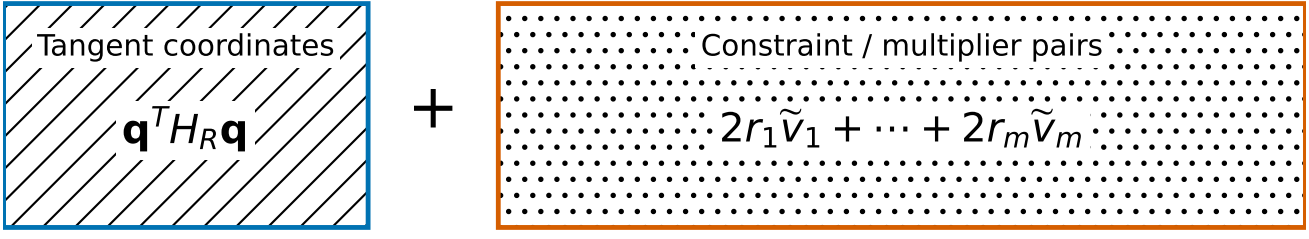

Invertible coordinates eliminate coupling: $\mathbf{v} = \tilde{\mathbf{v}} - G^T\mathbf{q} - \frac{1}{2}H_N\mathbf{r}$